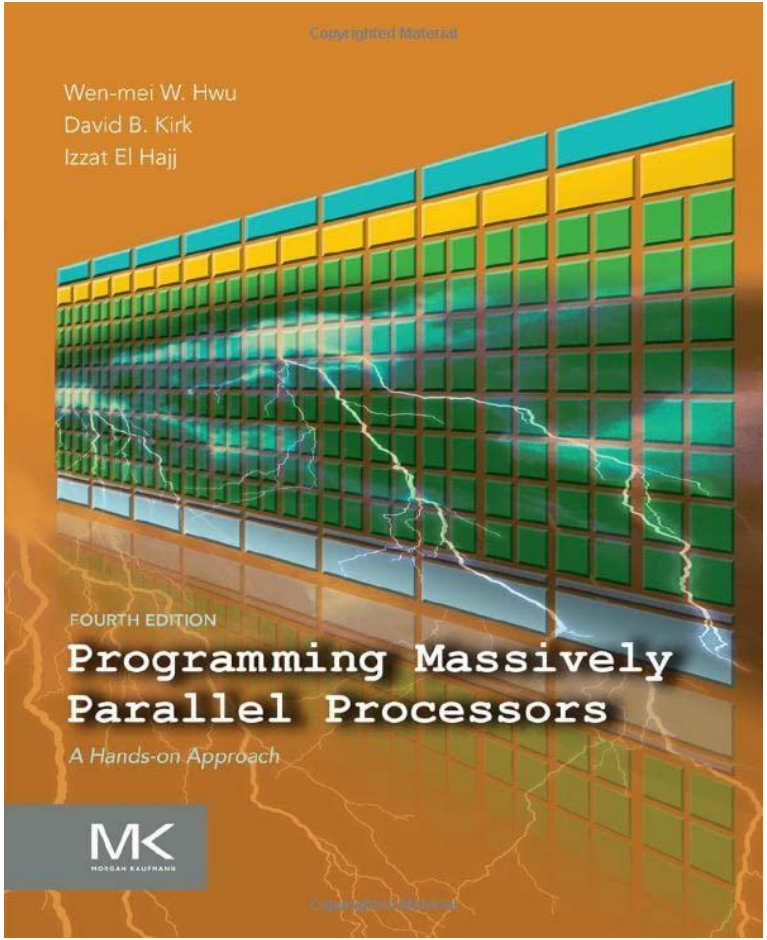


pmpp book ch. 1-3

CUDA-MODE

Lecture 2

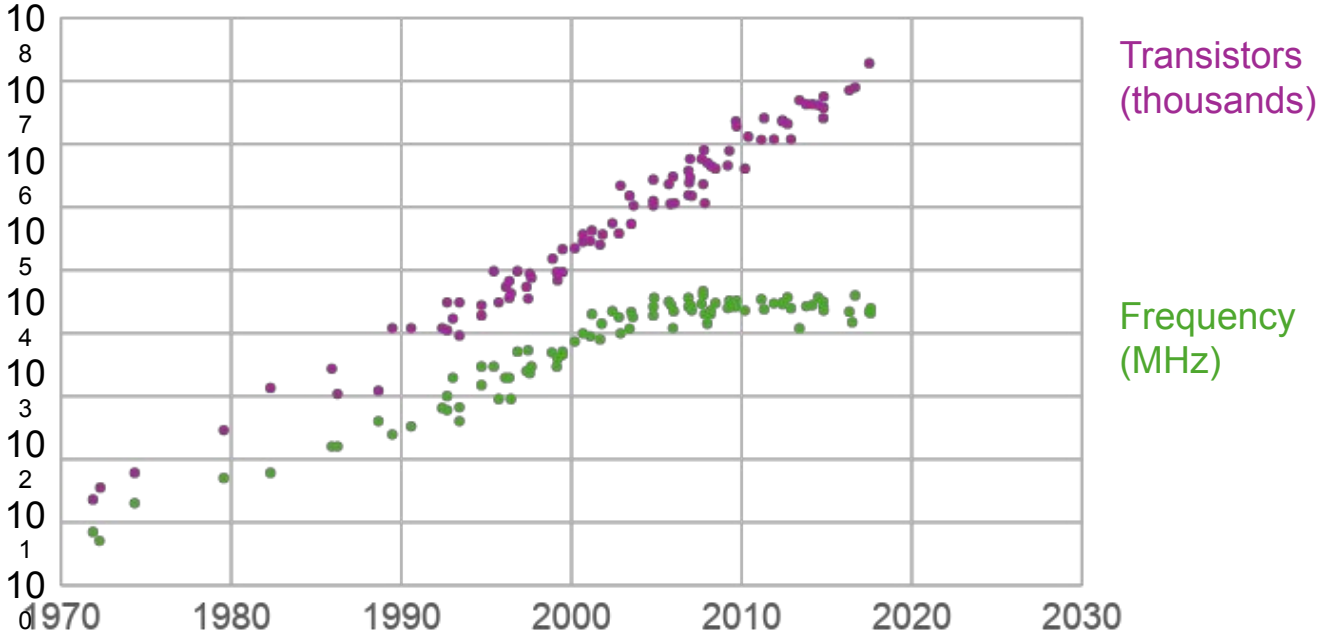
Agenda for Lecture 2

- 1: Introduction
- 2: Heterogeneous data parallel computing
- 3: Multidimensional grids and data

Ch 1: Introduction

- motivation: GPU go brrr, more FLOPS please
- Why? Simulation & world-models (games, weather, proteins, robotics)
- Bigger models are smarter -> AGI (prevent wars, fix climate, cure cancer)
- GPUs are the backbone of modern deep learning
- classic software: sequential programs
- higher clock rate trend for CPU slowed in 2003: energy consumption & heat dissipation
- multi-core CPU came up
- developers had to learn multi-threading (deadlocks, races etc.)

The Power Wall



Source: M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, C. Batten (1970-2010). K. Rupp (2010-2017).

(increasing frequency further would make the chip too hot to cool feasibly)

The rise of CUDA

- CUDA is all about parallel programs (modern software)
- GPUs have (much) higher peak FLOPS than multi-core CPUs
- main principle: divide work among threads
- GPUs focus on execution throughput of massive number of threads
- programs with few threads perform poorly on GPUs
- CPU+GPU: sequential parts on CPU, numerical intensive parts on GPU
- CUDA: Compute Unified Device Architect
- GPGPU: Before CUDA tricks were used to compute with graphics APIs (OpenGL or Direct3D)
- GPU programming is now attractive for developers (thanks to massive availability)

Amdahl's Law

- speedup = slow_sys_time / fast_sys_time
- achievable speedup is limited by the parallelizable portion ***p*** of programs

$$speedup < \frac{1}{1 - \boldsymbol{p}}$$

  - e.g., if ***p*** is 90%, speedup < 10×

- Fortunately, for many real applications, ***p*** > 99% especially for large datasets, and speedups >100× are attainable

Challenges

- "if you do not care about performance, parallel programming is very easy"
- designing parallel algorithms in practice harder than sequential algorithms
  e.g. parallelizing recurrent computations requires nonintuitive thinking (like prefix sum)
- speed is often limited by memory latency/throughput (memory bound)
- perf of parallel programs can vary dramatically based on input data characterists
- not all apps are "embarassingly parallel" - synchronization imposes overhead (waits)

Main Goals of the Book

1. Parallel programming & computational thinking
2. Correct & reliable: debugging function & performance
3. Scalability: regularize and localize memory access

- PMPP aims to build up the foundation for parallel programming in general
- GPUs as learning vehicle - techniques apply to other accelerators
- concepts are introduced hands-on as concrete CUDA examples

Ch 2: Heterogeneous data parallel computing

- heterogeneous: CPU + GPU
- data parallelism: break work down into computations that can be executed independently
- Two examples: vector addition & kernel to convert an RGB image to grayscale
- Independence: each RGB pixel can be converted individually
- L = r*0.21 + g*0.72 + b*0.07 (L=luminance)
- simple weighted sum

RGB->Grayscale, data independence

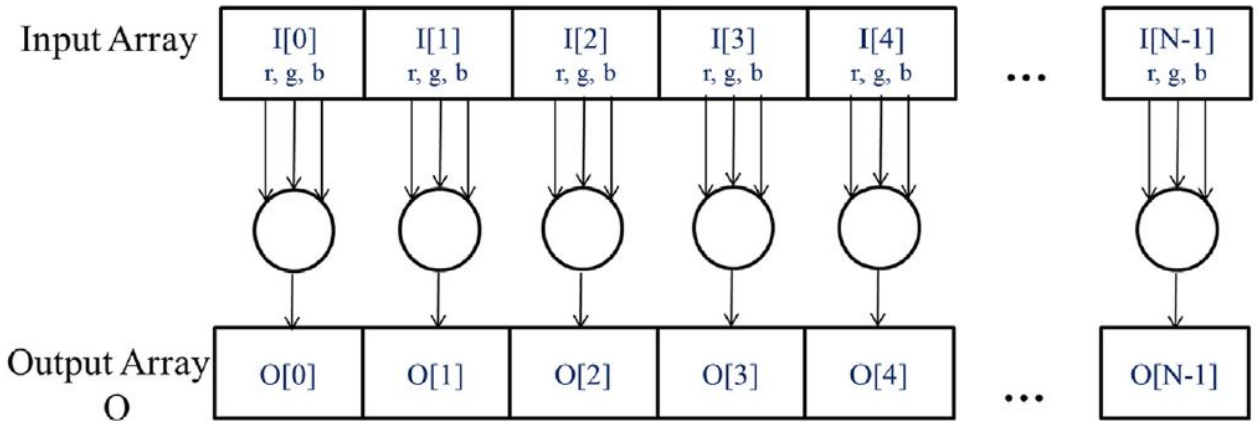


FIGURE 2.2

Data parallelism in image-to-grayscale conversion. Pixels can be calculated independently of each other.

CUDA C

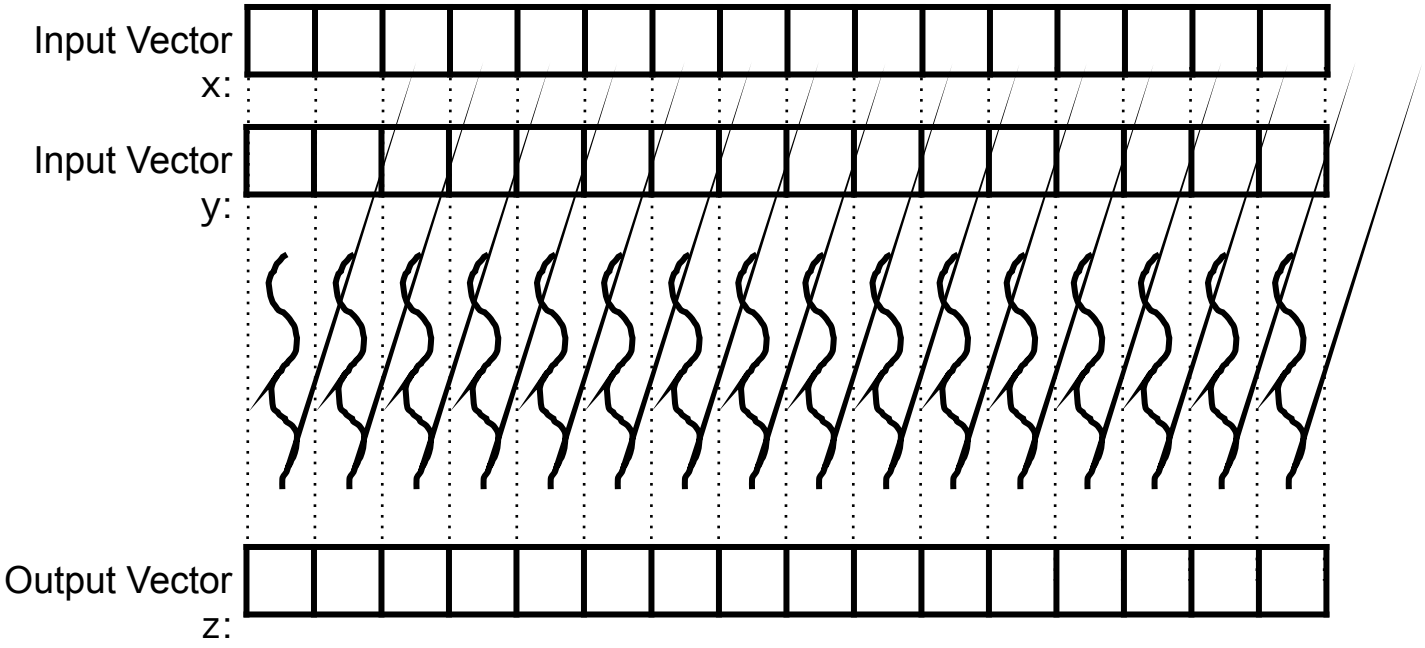
- extends ANSI C with minimal new syntax
- Terminology: CPU = host, GPU = device
- CUDA C source can be mixture of host & device code
- device code functions: kernels
- grid of threads: many threads are launched to execute a kernel
- CPU & GPU code runs concurrently (overlapped)
- on GPU: don't be afraid of launching many threads
- e.g. one thread pre (output) tensor element is fine

Example: Vector Addition

- vector addition example:
  - main concept loop -> threads
  - Easily parallelizable: all additions can be computed independently

- Naïve GPU vector addition:
  1. Allocate device memory for vectors
  2. Transfer inputs host -> device
  3. Launch kernel and perform additions
  4. Copy device -> host back
  5. Free device memory

- normally we keep data on the gpu as long as possible to asynchronously schedule many kernel launches



- One thread per vector element

CUDA Essentials: Memory allocation

- nvidia devices come with their own DRAM (device) global memory
  (in Ch 5 we learn about other mem types)

- `cudaMalloc` & `cudaFree`:

```
float *A_d;
size_t size = n * sizeof(float);      // size in bytes
cudaMalloc((void**)&A_d, size);  // pointer to pointer!
...
cudaFree(A_d);
```

cudaMemcpy: Host <-> Device Transfer

- Copy data from CPU memory to GPU memory and vice versa

```
// copy input vectors to device (host -> device)
cudaMemcpy(A_d, A_h, size, cudaMemcpyHostToDevice);
cudaMemcpy(B_d, B_h, size, cudaMemcpyHostToDevice);

...

// transfer result back to CPU memory (device -> host)
cudaMemcpy(C_h, C_d, size, cudaMemcpyDeviceToHost);
```

CUDA Error handling

- CUDA functions return `cudaError_t` .. if not `cudaSuccess` we have a problem ...
- always check returned error status ☺

Kernel functions fn<<<>>>

- Launching kernel = grid of threads is launched
- All threads execute the same code: Single program multiple-data (SPMD)
- Threads are hierarchically organized into **grid blocks** & **thread blocks**
- up to 1024 threads can be in a thread block

Kernel Coordinates

- built-in variables available inside the kernel: **blockIdx**, **threadIdx**
- these "coordinates" allow threads (all executing the same code) to identify what to do (e.g. which portion of the data to process)
- each thread can be uniquely identified by threadIdx & blockIdx
- telephone system analogy: think of `blockIdx` as the area code and `threadIdx` as the local phone number
- built-in `blockDim` tells us the number of threads in a block
- for vector addition we can calculate the array index of the thread
  `` `int i = blockIdx.x * blockDim.x + threadIdx.x;` ``

Threads execute the same kernel code

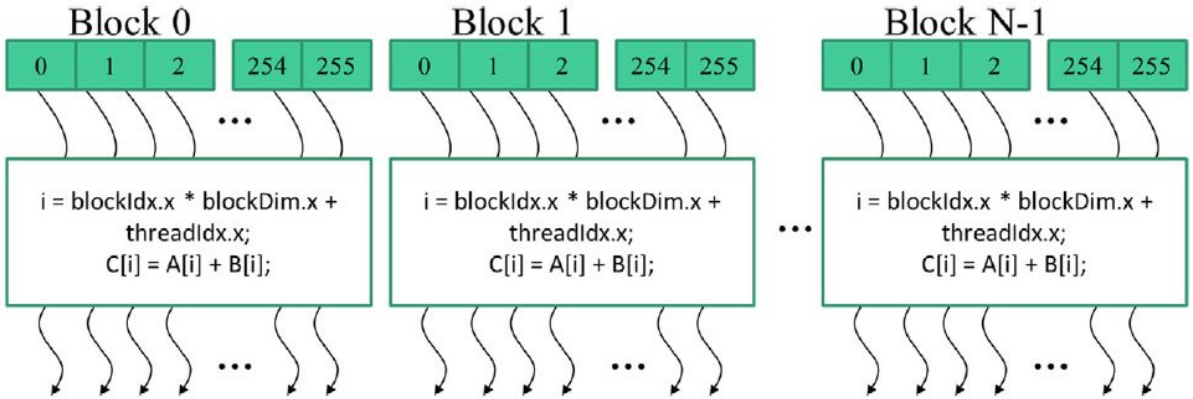


FIGURE 2.9

All threads in a grid execute the same kernel code.

__global__ & __host__

- declare a kernel function with `__global__`
- calling a `__global__` function -> launches new grid of cuda threads
- functions declared with `__device__` can be called from within a cuda thread
- if both `__host__` & `__device__` are used in a function declaration CPU & GPU versions will be compiled

Qualifier Keyword	Callable From	Executed On	Executed By
__host__ (default)	Host	Host	Caller host thread
__global__	Host (or Device)	Device	New grid of device threads
__device__	Device	Device	Caller device thread

FIGURE 2.11

CUDA C keywords for function declaration.

Vector Addition Example

- general strategy: replace loop by grid of threads!
- data sizes might not perfectly divisible by block sizes: always check bounds
- prevent threads of boundary block to read/write outside allocated memory

```
// compute vector sum C = A + B
// each thread peforms one pair-wise addition
__global__
void vecAddKernel(float* A, float* B, float* C, int n) {
 int i = threadIdx.x + blockDim.x * blockIdx.x;
 if (i < n) {   // check bounds
   C[i] = A[i] + B[i];
 }
}
```

Calling Kernels

- kernel configuration is specified between `<<<` and `>>>`
- number of blocks, number of threads in each block

```
dim3 numThreads(256);
dim3 numBlocks((n + numThreads - 1) / numThreads);
vecAddKernel<<<numBlocks, numThreads>>>(A_d, B_d, C_d,
n);
```

- we will learn about additional launch parameters (shared-mem size, cudaStream) later

Compiler

- nvcc (NVIDIA C compiler) is used to compile kernels into PTX
- Parallel Thread Execution (PTX) is a low-level VM & instruction set
- graphics driver translates PTX into executable binary code (SASS)

Ch 3: Multidimensional grids and data

- CUDA grid: 2 level hierarchy: **blocks**, **threads**
- Idea: map threads to multi-dimensional data
- all threads in a grid **execute the same kernel**
- threads in same block can access the same shared mem
- max block size: 1024 threads
- built-in 3D coordinates of a thread: **blockIdx**, **threadIdx** - identify which portion of the data to process
- shape of grid & blocks:
  - **gridDim**: number of blocks in the grid
  - **blockDim**: number of threads in a block

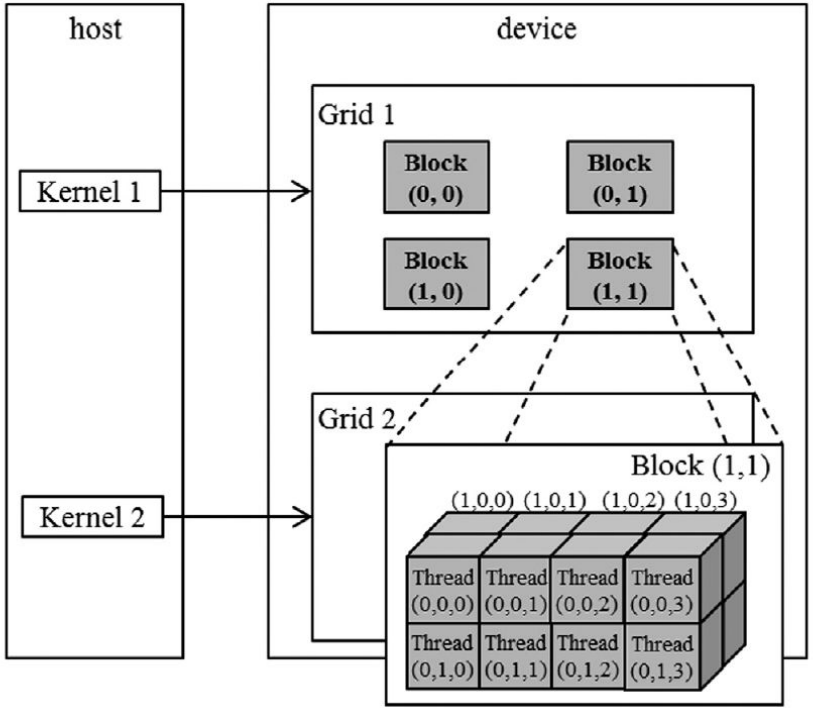


FIGURE 3.1

A multidimensional example of CUDA grid organization.

Grid continued

- grid can be different for each kernel launch, e.g. dependent on data shapes
- typical grids contain thousands to millions of threads
- simple strategy: one thread per output element (e.g. one thread per pixel, one thread per tensor element)
- **threads can be scheduled in any order**
- can use fewer than 3 dims (set others to 1)
- e.g. 1D for sequences, 2D for images etc.

```
dim3 grid(32, 1, 1);
dim3 block(128, 1, 1);
kernelFunction<<<grid, block>>>(..);
// Number of threads: 128 * 32=4096
```

Built-in Variables

- Built-in variables inside kernels:

```
blockIdx  // dim3 block coordinate
threadIdx // dim3 thread coordinate
blockDim  // number of threads in a block
gridDim       // number of blocks in a grid
```

- (blockDim & gridDim have the same values in all threads)

nd-Arrays in Memory

Actual layout in memory

0,0	0,1	0,2	0,3	1,0	1,1	1,2	1,3	2,0	2,1	2,2	2,3	3,0	3,1	3,2	3,3

Logical view of data

0,0	0,1	0,2	0,3
1,0	1,1	1,2	1,3
2,0	2,1	2,2	2,3
3,0	3,1	3,2	3,3

- memory of multi-dim arrays under the hood is flat 1d
- 2d array can be linearized different ways:

```
- row-major:
  A B C
  D E F
  G H I
```

```
- column-major:
  A D G
  B E H
  C F I
```

- torch tensors & numpy ndarrays use strides to specify how elements are laid out in memory.

Image blur example (3.3, p. 60)

- mean filter example blurKernel
- each thread writes one output element, reads multiple values
- single plane in book, can be extended easily to multi-channel
- shows row-major pixel memory access (in & out pointers)
- track of how many pixels values are summed
- handles boundary conditions in ln 5 & 25

FIGURE 3.6

An original image (*left*) and a blurred version (*right*).

Handling Boundary Conditions

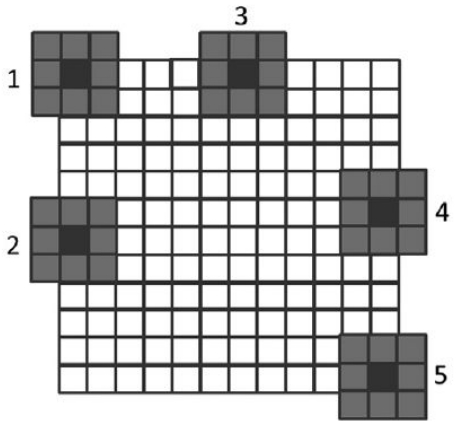


FIGURE 3.9

Handling boundary conditions for pixels near the edges of the image.

```
__global__
void mean_filter_kernel(unsigned char* output, unsigned char* input, int width, int height, int radius) {
    int col = blockIdx.x * blockDim.x + threadIdx.x;
    int row = blockIdx.y * blockDim.y + threadIdx.y;
    int channel = threadIdx.z;

    int baseOffset = channel * height * width;
    if (col < width && row < height) {

        int pixVal = 0;
        int pixels = 0;

        for (int blurRow=-radius; blurRow <= radius; blurRow += 1) {
            for (int blurCol=-radius; blurCol <= radius; blurCol += 1) {
                int curRow = row + blurRow;
                int curCol = col + blurCol;
                if (curRow >= 0 && curRow < height && curCol >=0 && curCol < width) {
                    pixVal += input[baseOffset + curRow * width + curCol];
                    pixels += 1;
                }
            }
        }

        output[baseOffset + row * width + col] = (unsigned char)(pixVal / pixels);
    }
}
```

Matrix Multiplication

- staple of science & engineering (and deep learning)
- compute inner-products of rows & columns
- Strategy: 1 thread per output matrix element
- Example: Multiplying square matrices (rows == cols)

```
__global__ void MatrixMulKernel(float* M, float* N,
                                float* P, int Width) {
    int row = blockIdx.y*blockDim.y+threadIdx.y;
    int col = blockIdx.x*blockDim.x+threadIdx.x;
    if ((row < Width) && (col < Width)) {
        float Pvalue = 0;
        for (int k = 0; k < Width; ++k) {
            Pvalue += M[row*Width+k]*N[k*Width+col];
        }
        P[row*Width+col] = Pvalue;
    }
}
```

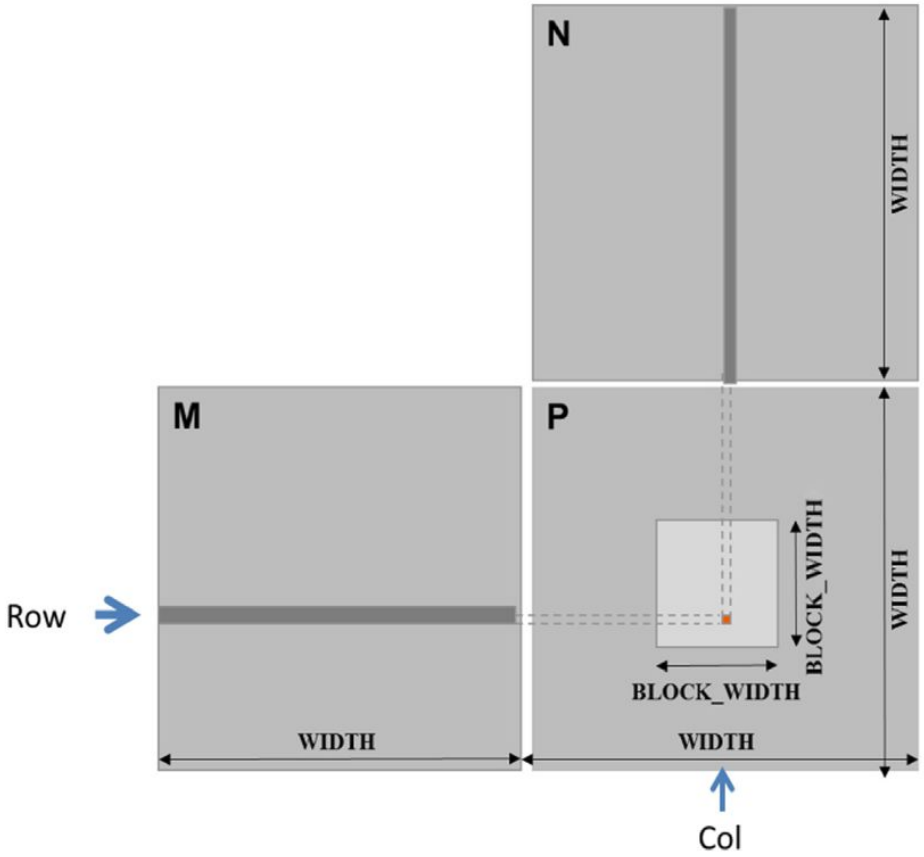


FIGURE 3.10

Matrix multiplication using multiple blocks by tiling P.